

The Utility Problem

Feeder voltage analysis is the process of measuring, analyzing, and interpreting data related to the voltage levels along the power distribution feeder, which is a conductor or a group of conductors that distribute power from a source, such as a substation, to a group of customers.

Visualizing data collected from voltage monitoring equipment is important because it allows power quality engineers and other personnel to quickly and easily identify patterns, trends, and anomalies in the voltage, which can help them to detect and diagnose problems in the electrical system. This can help to improve the overall power quality and reliability of the system and make the system more efficient.

Some of the benefits of visualizing voltage data include:

- Identifying patterns and trends - by visualizing data in a graphical and easy-to-understand format, such as charts, graphs, and diagrams, power quality engineers can quickly and easily identify patterns and trends in the data, which can help them to detect and diagnose problems in the electrical system.
- Identifying anomalies - by visualizing data in a graphical format, power quality engineers can quickly and easily identify anomalies in the voltage data, such as voltage sags, swells, transients, and harmonic distortion, which can help them to detect and diagnose power quality issues.
- Making informed decisions - by visualizing data in a graphical format, the utility can quickly and easily identify patterns, trends, and anomalies in the data, which can help the utility make informed decisions about how to improve power quality and reliability. This data can also be used by customers, helping to increase transparency.
- Improving the efficiency - visualizing data can help to improve the efficiency of the system, by identifying the areas where the system is not performing well and make the necessary improvements.

In summary, visualizing data collected from voltage monitoring equipment is an important aspect of maintaining and improving the overall power quality and reliability of an electrical system. It allows power quality engineers and other personnel to quickly and easily identify patterns, trends, and anomalies in the data, which can help them to detect and diagnose problems in the electrical grid.

The Data Solutions and Methods

Awesense Energy Transition Platform's Energy Data Model (EDM) is ready for such analysis. In general, two types of data are required:

1. The geospatial information system and connectivity data of the grid (GIS network model).
2. Time series data from measurement sensors in the grid for analysis.

Once the time series are ingested and/or integrated with the Awesense Platform, threshold values can be set for voltage. This is secured through True Grid Intelligence's (TGI) Grid Alerts feature.

The Awesense platform offers a very easy way to perform geospatial analysis of alerts and warnings in the grid once the alerts and warnings are generated and stored in the Energy data model. The unique integration of geospatial (GIS) data with time series data in the Awesense platform enables this type of analysis. Such geospatial analysis can be performed using the TGI Grid Alerts page or analysts can create their own analysis using the EDM API.

For building specialized analysis using the EDM APIs, the pre-built tracking functions speed up the creation of analyses for selected parts of the grid. The *grid_get_downstream()* function returns a list of *grid_elements* through which the current flows from the selected element. This makes it very easy to detect all the measurement sensors, for example, on a feeder or for a station or area. Alerts and warnings are assigned to sensors, and by combining these two data sources, we can therefore determine the number of alerts and warnings for each feeder, station or area in the grid. This data is then displayed on a map and in a table for easy interpretation. This is enabled by the trace and export functions based on the tracing that is part of the Awesense Platform. Without topology data, we would not be able to analyze smaller parts of the grid and get into the detail needed for such analyses.

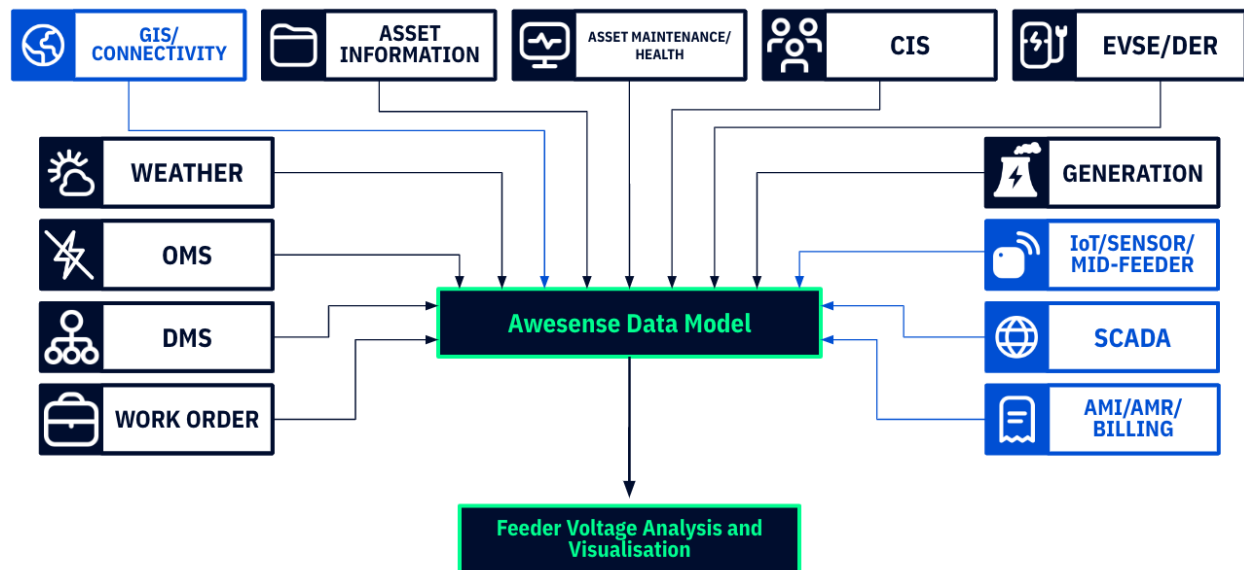
Feeder Voltage Analysis and Visualisation

[UC45-01] - Awesense Energy Transition Platform Use Case



Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. GitHub link to the Jupyter notebook can be provided upon request.